

Asymmetric Return-Volatility Relation, Volatility Transmission and Implied Volatility Indexes

Ihsan Ullah Badshah

Hanken School of Economics, Department of Finance and Statistics, P.O. Box 287,
FIN-65101 Vaasa, Finland. Phone: +358-6-3533 721, Fax: +358-6-3533 703,
Email: ibadshah@hanken.fi

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Abstract

The purpose of this study is twofold: First, to investigate the asymmetric return-volatility phenomenon with newly adapted robust volatility indexes VIX, VXN, VDAX and VSTOXX. Second, we examine the dynamic implied volatility transmissions across the implied volatility indexes using techniques such as Granger causality, generalized impulse response function and variance decomposition. We find pronounced negative and asymmetric return-volatility relationships between each volatility index and its corresponding stock market index. The VIX volatility index presents the highest asymmetric return-volatility relationship followed by the VSTOXX, VDAX and VXN volatility indexes, respectively. Moreover, there are significant spillover effects across the volatility indexes, bi-directional causality running between the volatility indexes. Further, in innovation accounting investigations, the VIX volatility index influences the other three volatility indexes considerably. However, in the European context, VDAX is the dominant source of information. Our findings have implications for trading strategies, hedging portfolios, pricing and hedging volatility derivatives, and risk management.

Keywords: implied volatility, index options, volatility indexes, volatility-return relation, volatility transmissions.

JEL Classifications: G12, G13, G15

1. Introduction

It is widely documented that implied volatility is superior to historical volatility when forecasting the future realized volatility of the underlying asset (e.g., Christensen and Prabhala, 1998; Fleming, 1998; Dumas et al., 1998; Blair et al., 2001). Implied volatility can be recovered by inverting the Black-Scholes (1973) formula. However, Britten-Jones and Neuberger (2000), and Jiang and Tian (2005) derived model-free (MF) implied volatility under the pure diffusion assumption and asset price processes with jumps. They showed that information content of MF volatility is superior to that of the Black-Scholes (BS) volatility: the MF implied volatility measure accounts for all strikes when computing implied volatility at a particular point in time, whereas the BS implied volatility measure is a point-based volatility and does not account for all strikes in computation, i.e., each strike has a separate implied volatility. Moreover, BS implied volatility is subject to both model and market efficiency, while MF volatility is only subject to market efficiency (see Poon and Granger, 2003). Thus, the major volatility indexes used to employ the BS at-the-money (ATM) implied volatility measure in their methodologies have now adopted MF volatility measures. The motives for adopting MF measures were: First, MF volatility index methodology is economically appealing and robust, as it accounts for both out-of-the-money (OTM) put and call options (i.e., volatility skew). Second, the previous volatility index measure (old S&P 100 volatility index, now called VXO) was upward biased, the bias being induced by trading day conversion, which is now omitted from the new VIX measure. Finally, with the new robust implied volatility index measure it is possible to replicate volatility derivatives (e.g., variance swaps) which was not possible with the previous measure.

As implied volatility is forward looking, that is, implied by market prices of options, and options are the common consensus of the market participants about the expected future volatility, implied volatility is the market expectation about the future realized volatility of

the underlying asset over the remaining life of an option. Similarly, the implied volatility index captures the market expectations. Major option exchanges, including the Chicago board of exchange (CBOE) and Deutsche Börse, have launched implied volatility indexes, robustly implying option information using MF volatility measures; for instance, the VIX implied volatility index for the S&P 500 stock index, VXN for the NASDAQ 100 stock index, VDAX for the DAX 30 stock index and VSTOXX for the Dow Jones (DJ) EURO STOXX 50 stock index. Thus, each implied volatility index is often referred to as the “investors fear gauge” (e.g., Whaley, 2000), because the level of the implied volatility index indicates the consensus view about the expected future realized stock index volatility. When the level of the volatility index increases, as a consequence, fear in the market increases; alternatively, when the level of the volatility index decreases, run-ups are triggered in the daily stock index prices. Additionally, the volatility index level indicates the degree of willingness of the market participants to pay in terms of volatility in order to hedge the downside risk of their portfolios with put options or long positions in call options with limited downside risks instead of positions in the underlying asset.¹

Likewise, the new volatility index measure corresponds to the option traders’ consensus opinions—options traders assumed to possess professional judgement—on the future direction of the volatility of the stock index for 30 calendar days. The new volatility index measure includes both OTM puts and calls and therefore moves with changes in options prices. For example, a negative or positive shock to the market induces adjustments in the hedging and trading strategies, consequently triggering changes in the prices of one type of option (i.e., put or call). The index volatility measure then moves in the direction of the market demand of a particular type of option and the underlying asset (see Bollen and Whaley, 2004). Hence, the new volatility index is an excellent tool for examining the

¹ See Simon (2003) for a detail discussion on the trading strategies.

relationship between market perception of the volatility and returns. Furthermore, this relation is asymmetric, implying that the volatility index reacts to negative and positive returns differently. Therefore, we believe that this negative asymmetric volatility-return relation should be more pronounced with the new robust volatility index measure (accounting for volatility skew) in contrast to the old volatility measure. A possible explanation for asymmetric volatility is that a put option is a downside hedging instrument and traders are always concerned about the downward moments in the market, traders always hedging their positions with OTM puts. Consequently, we find an asymmetric return-volatility relation (see, e.g., Bollen and Whaley, 2004). There are few well-known studies showing a significant negative and asymmetric relationship between returns and changes in the volatility index (e.g., Fleming et al., 1995; Whaley, 2000; Giot, 2005; Simon, 2003; Skiadopoulos, 2004; Low, 2004; Dennis et al, 2006; Hibbert et al., 2008). On the other hand, it is important to know whether implied volatility spills over from one market to another, i.e., market integration. However, the implied volatility spillover phenomenon across all markets has implications for risk managers, international portfolio managers and option traders. They need to characterize which market leads other markets and which market is the major source of implied information. More specifically, they need to know the volatility transmission from one market to another: How does a shock to one volatility index affect another volatility index? What is the magnitude and sign of the affect, and how long does the affect persist? Finally, how much shock of one market can explain the forecast error variance of another volatility index?

The first study on the relation between changes in the VIX index and S&P 100 index returns was conducted by Fleming et al. (1995). They investigated the time series properties of the old VIX (now VXO). They found a large negative contemporaneous correlation between VIX changes and stock index returns, the negative relationship showing significant

asymmetry. Another well-known study is that conducted by Whaley (2000), who examined the relationship of weekly changes in the VIX and S&P 100 returns. He documented that when VIX falls by 100 basis points, then the S&P 100 index increases by 0.469%, whereas when VIX increases by 100 basis points, the S&P 100 index falls by -0.707%. Thus, he found a large negative association of the changes in VIX and index returns with significant asymmetry, hence calling the VIX the “investor fear gauge”. Simon (2003) studied the VXN (NASDAQ volatility index) from January 1995 to May 2002, showing that VXN is inversely related to both positive and negative index returns. Furthermore, he found stable results across the bubble and post bubble periods. A more recent study by Hibbert et al. (2008) used a different approach to investigate the negative asymmetric return-volatility relation using the newly-developed VIX index. They found large negative and asymmetric association between VIX and stock index returns while incorporating both daily and intraday data. Furthermore, they showed, by quintile regression, that the new VIX measure can provide better explanation for this relation than the old VIX measure or employing a realized volatility measure.

There are many studies which have investigated volatility transmissions using historical volatilities (e.g., Hamao et al, 1990; Koutmos and Booth, 1995; Koutmos, 1996; Cifarelli and Paladino, 2005). However, we find few studies have examined the dynamic behaviour of implied volatility and volatility transmission across markets (see, e.g., Wagner and Szimayer, 2004; Nikkinen and Sahlström, 2004; Skiadopoulos, 2004; Nikkinen et al., 2006; Äijö, 2008). Äijö(2008) investigated the relation between the new European volatility indexes (VDAX, VSMI and VSTOXX). He found that the volatility indexes are highly correlated and indexes vary over time, the VDAX being the dominant source of information. VDAX Granger-causes both VSMI and VSTOXX, and the variance of the forecast errors of the implied volatility term structure of the VSTOXX and VSMI explain 65% and 35%, respectively, of the implied volatility term structure of the VDAX. Nikkinen and Sahlström (2004) studied international

equity market integration of the United States (US), United Kingdom (UK), German and Finnish markets with respect to implied volatility indexes using the VAR framework, testing for Granger causality, impulse responses and variance decompositions. Similarly, they found a high degree of integration among these markets: while the US market is the leading source of information transmitting to other markets generally, in the European context the German market leads other European markets, i.e., the UK and Finnish markets.

The purpose of this study is twofold: First, we investigate the asymmetric relationship between the returns of the stock market index and the changes in the implied volatility index: (1) to show the negative correlations in the stock market declines and run-ups; and (2) to characterize the market which presents the highest negative relationship, and how much this relationship is asymmetric. Second, we investigate the implied volatility transmission across markets, to establish the lead-lag relationship between the implied volatility indexes: How does a shock to one volatility index affect another volatility index? What is the magnitude and sign of the affect, and how long does the affect will persist? Finally, how much does one volatility index explain the forecast error variance of another volatility index for 10 days ahead? The first part of our study is related to the studies of Simon (2003) and Giot (2005) in terms of the volatility-return relationship. The former studied the relationship between the NASDAQ 100 index returns and changes in VXN, while the latter studied the relationship between S&P 100 and VIX, and between NASDAQ 100 and VXN. Both studies used the old volatility index measure. Our study differs from the above two cited studies in a number of ways. First, we incorporate the VIX, VXN, VDAX and VSTOXX volatility indexes and their corresponding stock indexes. Second, we consider the data on the newly-developed volatility indexes, i.e., based on the MF volatility robust measure, which incorporates the most information. Therefore, this asymmetric relationship should be more pronounced. The second part of our study, where we study the implied volatility transmission across markets, is

related to the studies on the volatility transmission by Nikkinen and Sahlström (2004) and Äijö (2008). The former studied the implied volatility transmission across US, UK, German and Finnish markets, while the latter studied volatility transmission between VDAX, VSTOXX and VSMI. Moreover, we study the implied volatility transmission across newly-developed volatility indexes (VIX, VXN, VDAX and VSTOX) using a recent data set.

Our main findings are that the volatility indexes VIX, VXN, VDAX and VSTOXX from February 2001 through June 2008 respond in a highly asymmetric fashion to the negative and positive returns of their corresponding stock indexes. Comparatively, the VIX volatility index presents the highest asymmetry followed by the VSTOXX, VDAX and VXN volatility indexes, respectively. These volatility indexes sharply rise in market turmoil and decline in market rallies. Moreover, there are considerable spillover effects across volatility indexes, bi-directional Granger causality running between the volatility indexes except between VIX and VXN. A negative shock to the VIX index surprisingly increases the levels of the other three volatility indexes—VXN, VDAX and VSTOXX—in a contemporaneous manner, the affects persisting for about 6 days. In the European context, the VDAX volatility index influences the VSTOXX volatility index. Therefore, a shock to VDAX increases the volatility level of VSTOXX. Conversely, a shock to VDAX affects the levels of the US volatility indexes, i.e., VIX and VXN. Moreover, in variance decompositions analysis, the VIX volatility index can explain the forecast error variances of 58%, 31% and 29% on average of VXN, VDAX and VSTOXX, respectively, while VDAX can explain 52% of the forecast error variance of VSTOXX.

This paper is organized as follows. Section 2 discusses the volatility indexes and their construction. Section 3 discusses the data. Section 4 examines the asymmetric return-volatility relation. Section 5 investigates the volatility transmissions. Section 6 summarizes and concludes.

2. Implied Volatility Indexes

2.1. *VIX and VXN*

The CBOE introduced a new VIX index in September 2003 based on options on the S&P 500 index. The index is determined from the bid and ask prices of the options underlying the S&P 500 index. The new VIX is independent of any option pricing model using the MF volatility measure. Thus, VIX provides an estimate of expected future realized stock market volatility for the 22 subsequent trading days (over 30 calendar days). However, the old VIX index, based on options on the S&P 100 index and introduced in 1993, has now moved to the new ticker symbol VXO. In contrast to the old VIX (now VXO) based on near the money BS implied volatilities options on the S&P 100 index, the new VIX uses market prices of options on the S&P 500 index.² This new MF VIX methodology accounts for both OTM put and call options (i.e., volatility skew). Thus, the new methodology is more appealing and robust. The CBOE's introduction of the new VIX was motivated by both theoretical and practical deliberations. First, the new VIX is economically more appealing because it is based on a portfolio of options, whereas the old VIX was based on the ATM option prices. Second, it is easy with the new VIX to replicate variance swap payoffs while using static positions in a range of options and dynamic positions in futures trading. Third, the new VIX has removed the induced upward bias of the old VIX in the trading day conversion (see, e.g., Carr and Wu, 2006). Similarly, in September 2003, the CBOE introduced the NASDAQ 100 volatility index (VXN) using the same MF methodology as that of VIX. CBOE calculated price histories for both VIX and VXN back to the years 1986, and 2001, respectively.

² The options on the S&P 500 index in comparison to the options on the S&P 100 index contain a much broader set of implied information, hence the new VIX is a more informative measure than the old VIX (now VXO).

2.2. VDAX and VSTOXX

The Deutsche Börse and Goldman Sachs jointly developed the methodology for the new VDAX and VSTOXX indexes. VDAX is based on options on the DAX 30 index whereas VSTOXX is based on options on the DJ Euro STOXX 50 index, consisting of the euro zone's 50 biggest blue-chip stocks. Options on DAX and DJ Euro STOXX 50 are traded on the derivatives exchange EUREX. The VDAX measure accounts for implied volatilities across all options of a given time to expiry (accounting for volatility skew). The methodology of VDAX, as for that of VIX, is also based on the MF volatility measure. However, the main VDAX volatility index is further based on eight constituent volatility indexes that expire in 1, 2, 3, 6, 9, 12, 18, and 24 months, respectively. The main VDAX is designed as a rolling index at a fixed 30 days to expiry via a linear interpolation of the two nearest of the eight available sub-indexes. The VDAX and its eight sub-indexes are updated every minute and therefore offer great advantages in terms of trading, hedging and introducing new derivatives on this index. The price histories for both VDAX and VSTOXX are calculated back to the years 1992 and 1999, respectively.

3. Data

This study employs data from four sources. We obtain the daily time series price data for the S&P 500 stock index, the NASDAQ 100 index, the DAX 30 index, and the DJ Euro STOXX 50 index from Thomson Financial DataStream. The data on the new VIX and VXN is obtained from the CBOE and the data on the new VDAX and VSTOXX is obtained from the Deutsche Börse and STOXX, respectively. The daily data for both stock and volatility indexes covers a period of seven years and four months, from February 2, 2001 to June 30, 2008, a total of 1933 trading days.

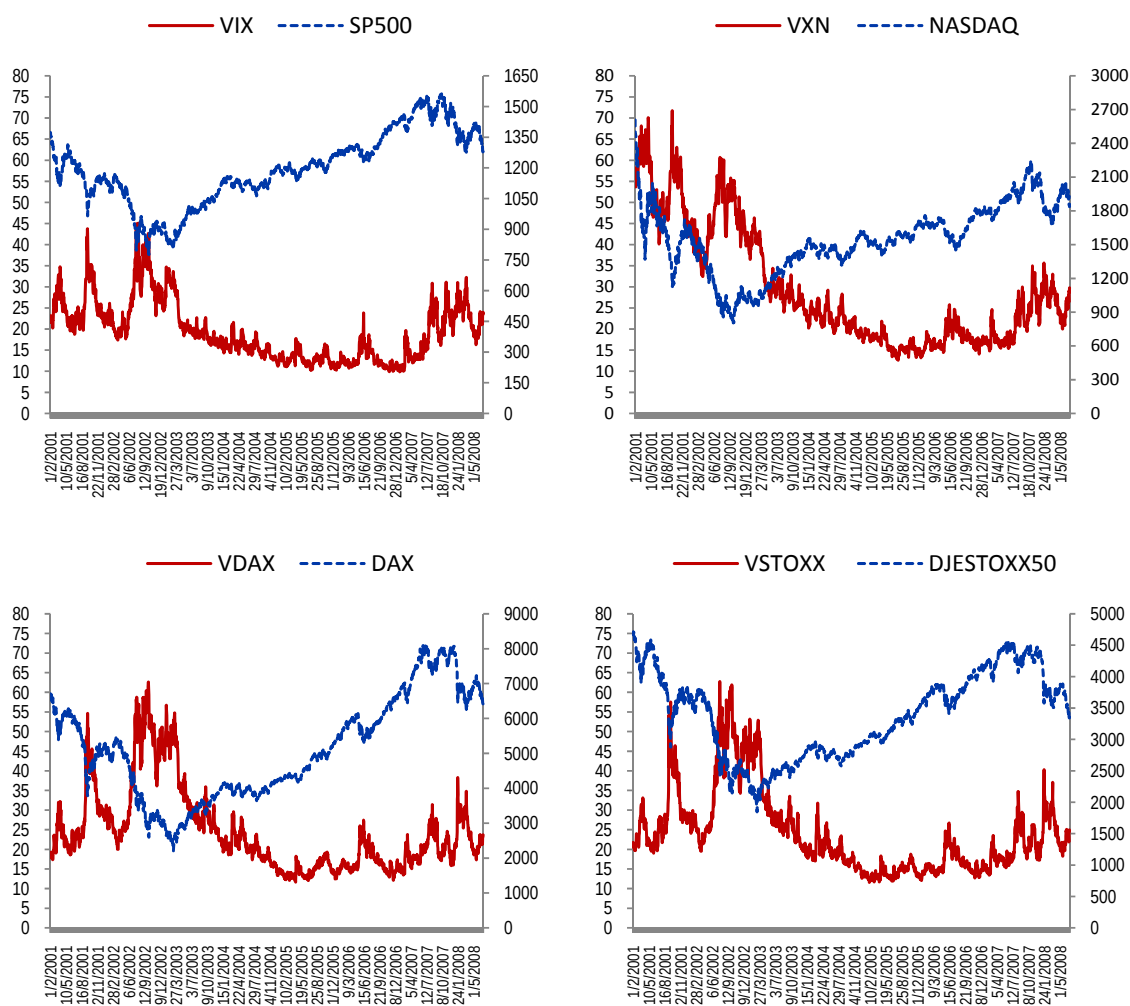


Fig. 1. Stock indexes versus implied volatility indexes from 2 February, 2001, to 30 June, 2008.

Fig.1 shows the daily closing levels (%) of the implied volatility indexes, i.e., VIX, VXN, VDAX and VSTOXX, and the corresponding stock market indexes (levels) from February 2, 2001 to June 30, 2008. Among the four volatility indexes, the VXN index presents the highest volatility level throughout our sample period, whereas the VIX shows the lowest volatility level. Similarly, the volatility indexes and stock indexes move inversely to each other. From the beginning of 2001 until the beginning of 2003, there are considerably high volatility levels. However, from 2004 onward until late 2007 we find upward movement in

the stock market indexes, whereas the corresponding volatility indexes are moving in the opposite direction to the stock markets, showing the lowest volatility levels. However, in the latter part of 2007 until June 2008, we again find somewhat high volatility levels, with the stock markets again moving downward because of the credit crunch crises and liquidity crunch crises.

Table 1 reports the summary statistics for the daily percentage continuously compounded returns of four stock indexes; the daily percentage changes in four implied volatility indexes; and tests for normality, autocorrelations and unit roots.

Table 1
Descriptive statistics.

	S&P500	NASDAQ	DAX	DJESTOXX50	VIX	VXN	VDAX	VSTOXX
Mean	-0.0036	-0.0181	-0.0022	-0.0175	0.0011	-0.0131	0.0025	0.0015
Median	0.0053	0.0177	0.0375	0.0000	-0.0300	-0.0200	-0.0200	-0.0500
Maximum	5.5732	10.2727	7.5526	7.0769	9.9200	10.5800	13.6900	19.1800
Minimum	-5.0467	-8.6138	-8.8746	-7.5929	-6.9900	-7.6000	-8.0500	-9.4200
Std. Dev.	1.0623	1.8268	1.5476	1.4177	1.2332	1.3331	1.4032	1.5387
Skewness	0.0500	0.0957	-0.14604	-0.0787	0.4737	0.2308	0.8794	1.4496
Kurtosis	5.6041	6.3885	6.6511	6.7562	9.4350	8.4480	12.378	20.277
Jarque-Bera	547.0	927.7	1080.5	1138.4	3407.4	2407.7	7333.0	24719.6
Prob	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ρ_1	-0.056*	-0.043	-0.048*	-0.036	-0.092**	-0.014	-0.013	-0.040
ADF	-46.45**	-45.85**	-46.07**	-45.52**	-35.02**	-33.32**	-44.52**	-45.73**
No. Obs	1933	1933	1933	1933	1933	1933	1933	1933

Table report the descriptive statistics of the stock market indexes and volatility indexes. The autocorrelation coefficients ρ , the Jarque-Bera and the Augmented Dicky Fuller(ADF) (an intercept is included in the test equation) test values are reported.

** and * Denote rejection of the null hypothesis at the 1% and 5% significance levels respectively.

The mean values for all eight stock index and volatility index series are not statistically different from zero. The tests for skewness and kurtosis confirm that the US stock indexes are positively skewed while the European stock indexes are negatively skewed, whereas all four volatility indexes are positively skewed as they should be. Furthermore, all eight series are

highly leptokurtic with respect to normal distribution. Likewise, the Jarque-Bera statistics rejects normality for each of the stock index and volatility index series. The first autocorrelation coefficients show that changes in the VIX series present strong autocorrelations. Autocorrelations in the return series of S&P 500 and DAX 30 are also evident, consequently confirming mean-reversion property. We investigate the stationarity in all eight series (i.e., stock and volatility indexes) by applying the augmented Dicky Fuller (ADF) test of unit root. The results in Table 1 show the rejection of unit roots in each series at the 1% significance level. Hence, all eight series are stationary.

4. Return-Volatility Relation

4.1. Correlation Analysis

We begin by estimating the time series monthly correlations between returns on the stock indexes and the changes in the volatility indexes from February 2, 2001, to June 30, 2008. The time series correlations are translated into four graphs in Fig.2. The monthly correlations between the S&P 500 index return and the changes in VIX are shown in the subplot at the top left of Fig.2. There are a number of noticeable findings: First, we find negative correlation large at -0.79 on average. The lowest correlation level is in the second half of 2001 at -0.94; the highest correlation level is in the first half of 2003 at -0.37.

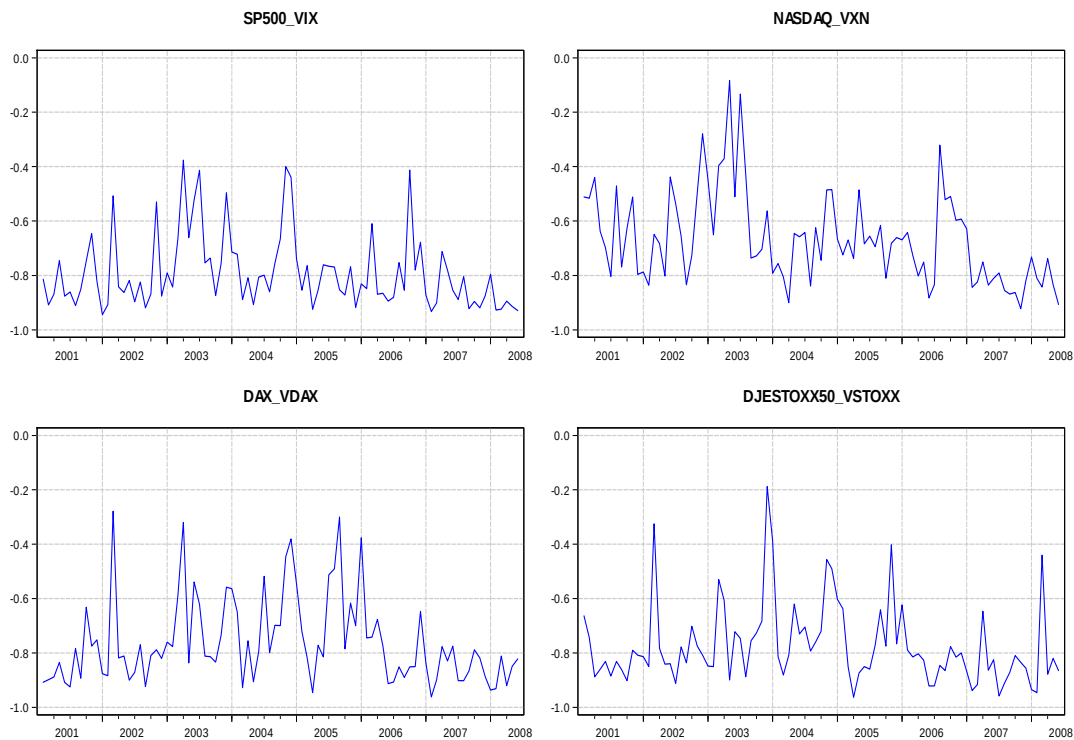


Fig. 2. Monthly correlations between stock indexes and volatility indexes from 2 February, 2001, to 30 June, 2008.

Second, monthly correlations between the NASDAQ index returns and changes in VXN are

plotted in the subplot at the top right of Fig.2. The correlations clearly vary greatly during our sample period, although the correlation is negative and large at -0.66 on average. The lowest level is reached in the first half of 2004 at -0.92; the highest level reached is in the first half of 2003 at -0.08. Third, the correlations between the DAX 30 index returns and the changes in VDAX are shown in the subplot at the bottom left of Fig.2. Here the average correlation is negative, -0.76. The lowest level reached in our sample period is in the first half of 2007 at -0.96. The highest level reached is in the first half of 2002 at -0.27. Finally, looking at the correlations between the DJ Euro STOXX 50 index returns and the changes in VSTOXX, there are similar patterns as those of the German market. The average correlation is -0.76. The lowest level reached is in the first half of 2005 at -0.96; the highest level reached is in the second half of 2003 at -0.18.

Thus, the correlation analysis in Fig.2 provides very important findings: First, we assert that there is noticeable evidence that the stock index return is highly negatively correlated to the changes in the volatility indexes both on the US and European markets. The correlations are pronounced because the newly adopted volatility index measures are economically robust, accounting for the volatility skew. However, these would not be clear with the old volatility index measures. Second, the negative correlation increases as the market declines, evident during 2001-2002, during which the internet bubble crises and burst of the bubble occurred, and 2008, when the credit crunch crises began. Thus, these findings lead us to study further the contemporaneous volatility-return relation and the asymmetries.

4.2. Asymmetric Return-Volatility Relation

Previous research found a significant negative relationship between stock index returns and the changes in the implied volatility index (see, e.g., Whaley, 2000; Simon, 2003; Skiadopoulos, 2004; Giot, 2005; Hibbert et al., 2008). However, our study considers the

newly-developed implied volatility indexes, based on robust MF implied volatility measure, instead of the old implied volatility indexes, based on the ATM-BS implied volatility measure, as we believe that there should be a more visible asymmetric return-volatility relation with the new volatility indexes in contrast to the previous documented evidence that considers the old volatility indexes.³ The new volatility index measure is economically appealing and robust, as it accounts for volatility skew. Volatility skew is an obvious phenomenon previously documented by many researchers, which is important to capture but was not accounted for in the old volatility index measure. A well-known study is that by Bollen and Whaley (2004), who investigated the relationship between net buying pressure and the shape of the implied volatility function (IVF) for both index and individual stock options. Their results show that buying pressure of S&P 500 index put options affects the changes in the implied volatility of S&P 500 options considerably, and, call option demand affects the changes in the implied volatility of individual stock options. They assert that when the buying pressure of index put options increases, particularly from institutional investors seeking to hedge their portfolios, the ability of arbitrageurs to bring back the price into alignment is limited. Consequently, this buying pressure of index put options drives the sloping shape of the IVF permanently downward. Furthermore, they simulate a trading strategy of selling options and delta-hedging the positions with the underlying index. This generates significant profit, particularly for the moneyness category that contains the OTM put options, i.e., the demand for portfolio insurance.

Thus, the new volatility index changes with the changes in option prices. As the new volatility index accounts for both OTM put and call types of options, a shock to the market that leads to a change in the prices of the one type of option in contrast to the other type

³ Earlier Black (1976) and Christie (1982) attributed this asymmetric phenomenon to leverage effect. However, French et al. (1987), Schwert (1989, 1990) suggest that leverage ratio alone is unable to explain this asymmetric phenomenon.

results in an adjustment to the volatility index following a similar direction as the net change. Nevertheless, the new volatility index is implied from both fear and exuberance imbedded in the option prices. Moreover, the majority of option market traders are very informed and highly skilled. Hence, the new volatility index is a good candidate to examine this asymmetric volatility-return relationship.

To assess the contemporaneous asymmetric relationship between the returns on the stock index and changes in the volatility index, we therefore follow a similar methodology as that of Simon (2003) and Giot (2005).⁴ Previously this asymmetric relationship was confirmed by the studies of Simon (2003) and Giot (2005). We regress the daily changes in the volatility index (denoted ΔVI_t^i , where $i = \Delta VIX, \Delta VXN, \Delta VDAX, \Delta VSTOX$) on the daily percentage continuously compounded returns of the stock market index (denoted R_t^j , where $j = \text{S\&P 500, NASDAQ, DAX, DJ Euro STOXX 50}$), variable $R_t^{+,j}$ for positive returns and variable $R_t^{-,j}$ for negative returns. For the positive returns $R_t^{+,j} = R_t^j$ if $R_t^j > 0$, and $R_t^{+,j} = 0$, otherwise. On the other hand, for the negative returns $R_t^{-,j} = R_t^j$ if $R_t^j < 0$, and $R_t^{-,j} = 0$, otherwise.

$$\Delta VI_t^i = a_0 + a_1 \Delta VI_{t-1}^i + a_2 R_t^{+,j} + a_3 R_t^{-,j} + u_t \quad (1)$$

The model is estimated for the contemporaneous relationship between the relative percentage changes in each implied volatility index and its corresponding stock index for the sample period from January 2, 2001 through June 30, 2008. Newey-West (1987) correction was used to correct for heteroscedasticity and autocorrelation of the estimates shown in Table 2. When we look at the coefficients of variables R_t^+ and R_t^- in Rows 3 and 4, respectively, from the absolute difference, it is apparent that there are asymmetric effects for all four estimations. The absolute values of R_t^- are higher than the absolute values of R_t^+ . Moreover, Wald-test

⁴ The relationship would behave differently for negative and positive stock index returns.

for coefficients is applied in order to find the statistical difference between the coefficients a_2 and a_3 in equation-1. The null hypothesis (i.e., the coefficients for negative and positive returns are equal) for the Wald-test is significantly rejected for each of the four regressions. These results imply the asymmetric return-volatility relation; that the negative returns for the stock indexes are linked to much higher volatilities for the implied volatility indexes VIX, VXN, VDAX and VSTOXX, than are positive returns.

Table 2

Contemporaneous relative changes in volatility index versus stock market index(daily returns).

Variable	Dependent variable: ΔVIX_t Coeff. (t-statistics)	Dependent Variable: ΔVXN_t Coeff. (t-statistics)	Dependent variable: $\Delta VDAX_t$ Coeff. (t-statistics)	Dependent variable: $\Delta VSTOXX_t$ Coeff. (t-statistics)
a_0	-0.1076* (-4.241)	-0.0951** (-2.712)	-0.1361** (-3.657)	-0.1830** (-3.785)
ΔVI_{t-1}	-0.0640** (-3.439)	0.0082 (0.355)	0.0279 (1.431)	-0.0243 (-0.848)
R_t^+	-0.7918** (-16.000)	-0.3959** (-9.558)	-0.5788** (-13.770)	-0.6663** (-11.952)
R_t^-	-1.0727** (-21.503)	-0.5128** (-13.189)	-0.8346** (-16.107)	-1.0129** (-13.085)
Adj. R^2	0.6626	0.3913	0.6271	0.6223
No.Obs.	1933	1933	1933	1933

The table report results from the regression of each implied volatility index on a set of variables: The following specification is estimated for each volatility index separately: $\Delta VI_t^i = a_0 + a_1 \Delta VI_{t-1}^i + a_2 R_t^{+,j} + a_3 R_t^{-,j} + u_t$, where ΔVI_t^i is the daily changes in volatility index, ΔVI_{t-1}^i AR(1) term of volatility index, R_t^+ the positive stock index return, R_t^- the negative stock index return. Furthermore, the model is estimated with Newey-West correction for heteroscedasticity and autocorrelation. ** and * Denote rejection of the null hypothesis at the 1% and 5% significance levels, respectively.

More specifically, looking at Column 2 of Table 2, the results indicate that the impact of the negative and positive S&P 500 index returns on the VIX is highly asymmetric, both

coefficients being statistically significant at the 1% significance level. The coefficient estimates for the R_t^- variable imply that a 1% decline in S&P 500 returns is linked to a 1.073% increase in the VIX level, whereas the coefficient estimates for the R_t^+ variable imply that a 1% increase in S&P 500 returns is linked to a 0.792% decrease in the VIX level. Column 3 of Table 2 shows strong asymmetric patterns for both NASDAQ and VXN. The coefficient estimates $R_t^-(R_t^+)$ suggest that a 1% decline (increase) in NASDAQ returns is associated with a 0.513% (0.396%) increase (decrease) in the VXN index. The coefficients are statistically significant at the 1% significance level. Column 4 of Table 2 also shows an asymmetric impact for DAX and VDAX. The coefficient estimates $R_t^-(R_t^+)$ imply that a 1% decline (increase) in the DAX returns is associated with a 0.835% (0.579%) increase (decrease) in the VDAX index, the coefficients being statistically significant at the 1% significance level. Finally, Column 5 of Table 2 show results for DJ Euro STOXX 50 and VSTOXX: the coefficient estimates $R_t^-(R_t^+)$ imply that a 1% decline (increase) in the DJ Euro STOXX 50 returns is associated with a 1.013% (0.666%) increase (decrease) in the VSTOXX index, the coefficients being statistically significant at the 1% significance level.

The main conclusion drawn from Table 2 is that the negative and positive stock index returns trigger the implied volatility index to move in completely the opposite direction and in an asymmetric fashion, i.e., negative returns have much greater impact on volatility index than positive returns.⁵ Moreover, the VIX and VSTOXX volatility indexes present greater asymmetric patterns in comparison with the VDAX and VXN volatility indexes. As new volatility indexes account for OTM puts, the asymmetry is very pronounced for each volatility index. Because investors hedge their downside risk by taking positions in the OTM put options, in market turmoil there is greater buying demand for put than call options, which

⁵ Our results are consistent with those of Simon (2003).

leads to higher volatilities than in market rallies. Consequently, negative stock index returns induce an increase in the levels of the volatility indexes. Our results are consistent with those of Boolean and Whaley (2004).

5. Implied Volatility Transmission

5.1. Correlation Analysis

We compute time series monthly correlations between the changes in the volatility indexes VIX, VXN, VDAX and VSTOXX from February 2, 2001 through June 30, 2008. The monthly correlations are presented graphically in Fig.3. There are a number of noteworthy results: First, the four volatility indexes are positively correlated. The US volatility indexes—VIX and VXN—demonstrate high correlations throughout the sample period, the mean correlation being 0.76. Similarly, the European volatility indexes—VDAX and VSTOXX—show considerably high correlations, the mean correlation being 0.88. Moreover, the volatility indexes across Atlantic, between US and Europe, present relatively low but positive correlations. The average correlation for VIX and VDAX is 0.48; VIX and VSTOXX, 0.46; VXN and VDAX, 0.45; and VXN and VSTOXX, 0.43, the least correlation among the four volatility indexes.

Second, the time series behaviour of the correlations is of note. For example, the correlation between the regional indexes presents less variation, although there are high variations across regions. VIX and VXN correlation reaches the highest in the first half of 2007 at 0.98; the lowest level reached is in the first half of the 2005 at 0.38. While the highest correlation of VIX and VDAX is reached in the second half of 2003 at 0.83, the lowest level is reached in the second half of 2005 at -0.26, interestingly negative. Similarly, the VIX and VSTOXX correlation reaches the highest level in the second half of 2002 at 0.85, while it reaches the lowest level in the second half of 2003 at -0.16, also negative. Finally, we can say that

markets in the same region are highly integrated, and markets across regions also present integration.

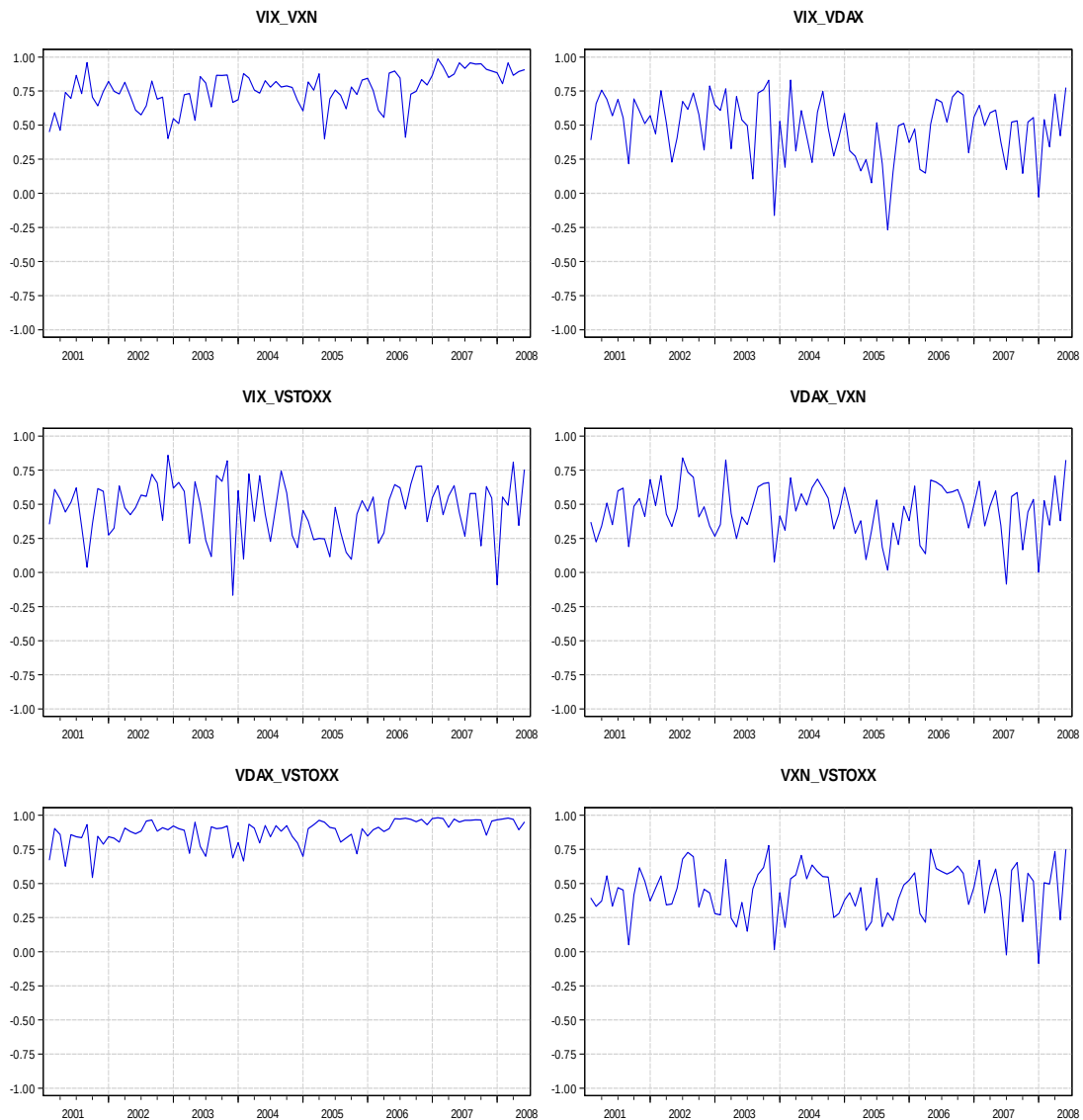


Fig. 3. Monthly correlations between volatility indexes from 2 February, 2001, to 30 June, 2008.

Thus, the results of the correlation analysis in Fig.3 are highly interesting: First, we assert that there is visible evidence that the new volatility indexes are considerably positively correlated both locally, i.e. within US and Europe, and across the Atlantic. Second, increase in positive correlations during market declines are evident during 2001-2002, the time of the

internet bubble crises, and latter part of 2008, the beginning of the credit crunch. These findings motivate us to explore the relation between these volatility indexes more thoroughly.

5.2. VAR Model for Implied Volatility Indexes

To investigate the implied volatility transmissions across the major volatility indexes, VIX, VXN, VDAX and VSTOXX, we begin with the Granger (1999) causality test and vector autoregressive (VAR) model. The VAR model was developed by Sims (1980) to investigate the dynamic impact of random innovations on a system of variables. This model successfully treats each endogenous variable in a system as a function of the lagged value of all the endogenous variables in the dynamic simultaneous equations system. Hence, the VAR model can be represented mathematically as follows:

$$\Delta VI_t = \alpha + \sum_{i=1}^L \Phi_i \Delta VI_{t-i} + u_t, \quad (2)$$

where $\Delta VI_t = (\Delta VIX_t, \Delta VXN_t, \Delta VDAX_t, \Delta STOXX_t)'$ is an $m \times 1$ vector of endogenous variables representing the daily percentage changes in the volatility indexes; $\{\Phi_i, i = 1, 2, 3, \dots, L\}$ is an $m \times m$ matrix of coefficients; and u_t is an $m \times 1$ vector of innovations which can be contemporaneously correlated, however, uncorrelated with its own lag values and with other variables. Thus, the VAR model is estimated by the ordinary least squares (OLS) estimation method. However, in order to select an appropriate number of lags for the dynamic VAR system, four different selection criteria are used: the Akaike information criterion (AIC); Schwartz information criterion (SIC); Hannan-Quinn information criterion (HQIC); and the likelihood ratio (LR) test.

Table 3 reports AIC, HQIC, SIC and LR lag order selections: AIC suggests lag=11, HQIC suggests lag=3, LR suggests lag=12, and SIC suggests lag=2. Hence, the parsimonious SIC

with lag=2 is chosen, and consequently, analysis with Equation 2 is conducted with lag=2.

Table 3
VAR lag order selection criteria.

Lag	AIC	HQIC	LR	SIC
0	11.14797	11.15223		11.15955
1	10.88439	10.90569	536.9346	10.94229
2	10.79907	10.83741	194.9846	10.90328*
3	10.77620	10.83158*	75.42823	10.92672
4	10.76165	10.83407	59.41740	10.95848
5	10.74546	10.83493	62.39998	10.98861
6	10.73138	10.83788	58.28639	11.02084
7	10.72903	10.85257	35.96031	11.06481
8	10.71136	10.85194	64.80923	11.09345
9	10.70871	10.86634	36.37254	11.13712
10	10.69518	10.86984	56.75997	11.16990
11	10.69421*	10.88591	33.07276	11.21524
12	10.69606	10.90481	27.71474*	11.26341

*Denote lag order selected by each criterion
AIC: Akaike Information Criterion
HQIC: Hannan-Quinn Information Criterion
LR: sequential modified LR test statistic [each test at 5% level]
SIC: Schwarz Information Criterion

To study the direction of the causality between the volatility indexes, we apply the Granger causality test, which establishes the causal relationship between the volatility variables and accordingly confirms the lead-lag relation. Granger causality establishes that volatility index *i* is Granger-caused by volatility index *j* if the information in the past and present values of volatility index *j* helps to improve the forecasts of volatility index *i*. The results for VIX, VXN, VDAX and VSTOXX are reported in Table 4 for lags 2 and 3 with corresponding values of the F-tests. As can be seen at lag 2, bi-directional causality is at work between most of the indexes except between VIX and VXN, which do not significantly cause one other. However, at lag=3, similarly, bi-directional causality is observed but now VIX Granger-causes significantly VXN, but not vice versa.

Table 4

Granger causality for implied volatility indexes.

Null Hypothesis	2 lags		3 lags	
	<i>F</i> -Statistics	<i>P</i> -Value	<i>F</i> -Statistics	<i>P</i> -Value
ΔV_XN does not Granger Cause ΔVIX	0.7851	0.456	1.5980	0.187
ΔVIX does not Granger Cause ΔV_XN	2.0839	0.124	4.3631**	0.004
$\Delta VDAX$ does not Granger Cause ΔVIX	4.0628*	0.017	4.8836**	0.002
ΔVIX does not Granger Cause $\Delta VDAX$	72.367**	0.000	52.6115**	0.000
$\Delta VSTOXX$ does not Granger Cause ΔVIX	3.4713*	0.031	5.3736**	0.001
ΔVIX does not Granger Cause $\Delta VSTOXX$	101.38**	0.000	70.6272**	0.000
$\Delta VDAX$ does not Granger Cause ΔV_XN	4.8816**	0.007	3.8500**	0.009
ΔV_XN does not Granger Cause $\Delta VDAX$	23.165**	0.000	16.2836**	0.000
$\Delta VSTOXX$ does not Granger Cause ΔV_XN	5.70414**	0.003	4.8418**	0.002
ΔV_XN does not Granger Cause $\Delta VSTOXX$	40.118**	0.000	27.1523**	0.000
$\Delta VSTOXX$ does not Granger Cause $\Delta VDAX$	6.2354**	0.002	6.7561**	0.000
$\Delta VDAX$ does not Granger Cause $\Delta VSTOXX$	7.0487**	0.000	6.3475**	0.000

** and * Denote rejection of the null hypothesis at the 1% and 5% significance levels respectively.

5.2.1. Generalized Impulse Response Functions

Granger causality in a VAR system using the F-test provides information about which variables have impact on the future values of each of the variables in the VAR system. However, F-test values do not provide the sign of the relationship, speed or persistence (see, e.g., Brooks, 2002). This information can be obtained by investigation of the VAR's impulse responses and variance decompositions.

An impulse response function measures the responses of the variables—in our case VIX, V_XN, VDAX and VSTOXX—in the dynamic VAR system to a shock to each variable. That is, a one standard error shock is applied to the error of a variable, and the effect on the dynamical VAR system over a specified period of time is recorded.

After estimating the parameters of the VIX, V_XN, VDAX and VSTOXX variables in

Equation (2) consistently by OLS, we then turn to estimate the generalized impulse response functions using the approach of Pesaran and Shin (1998). Generalized impulse response analysis is advantageous since it does not require orthogonalization of shocks and is invariant to the reordering of the variables in the VAR system.⁶ Mathematical illustration of impulse response functions is as follows:

The moving average representation of the Equation (2) is

$$z_t = \omega + \sum_{i=1}^{\infty} \Psi_i u_{t-i}, \quad (3)$$

where z_t is an m -variate stochastic process; ω is the deterministic part of z_t , and the $m \times m$ coefficient matrices Ψ_i is computed using the following looping relations:

$$\Psi_i = \sum_{p=1}^i B_p \Psi_{i-p}, \quad i = 1, 2, \dots, \quad (4)$$

where $\Psi_0 = I_m$ and $\Psi_i = 0$ for $i < 0$, and $G_i = \Psi_i \theta$.

Thus, the impulse response function computes at a given time the time profile of the effect of shocks on future values of the volatility index in a dynamic system.

Here Ψ_i matrices are the dynamic multipliers of the system, representing the model's response to a unit shock in each of the volatility indexes. The response of z_i to a unit shock in z_j is thus specified by the following sequence, called the impulse response function:

$$\Psi_{ij,1}, \Psi_{ij,2}, \Psi_{ij,3} \dots, \quad (5)$$

where $\Psi_{ij,k}$ is the ij th element of matrix Ψ_k . Since we know that $E(u_t u_t') = \Sigma$, the components of u_t are contemporaneously correlated. Pesaran and Shin (1998) suggest that whenever these correlations are high then, separating the response of z_i to z_j from its response to other shocks that are correlated with u_{jt} is not possible and the resulting values

⁶ Same notations as of Pesaran and Shin (1998) are used.

by simulation will be misleading. Thus, generalized impulse responses should be considered instead. The generalized impulse response function shocks only one element of u_t , say the j th element, and integrates out the effects of distribution of the errors of the other shocks. In our case we have

$$GI_z(n, \rho_j, \Omega_{t-1}) = E(z_{t+n} | u_{jt} = \rho_j, \Omega_{t-1}) - E(z_{t+n} | \Omega_{t-1}), \quad (6)$$

assuming that u_t has a multivariate normal distribution. However, for non-normal error distribution, one needs to use stochastic simulation to obtain the conditional expectations $E(u_t | u_{jt} = \rho_j)$ (see, e.g., Koop et al., 1996; Pesaran and Shin, 1998; Mills, 1999). As our data shows non-normality, we therefore use the generalized impulse response function in combination with stochastic simulations thus:

$$E(u_t | u_{jt} = \rho_j) = (v_{1j}, v_{2j}, \dots, v_{ij})' v_{jj}^{-1} \rho_j = \sum e_j v_{jj}^{-1} \rho_j. \quad (7)$$

Hence, the generalized impulse response measures the impact of a one standard error shock in scaled form at time t on future values of z at time $t+n$:

$$\Psi_j^g(n) = v_{jj}^{-1/2} \Psi_n \sum e_j, \quad n = 0, 1, 2, \dots \quad (8)$$

Table 5 reports the accumulated generalized impulse responses of VIX, VXN, VDAX and VSTOXX to a unit step shock (one standard unit) in the innovations of each respective volatility index, i.e., the same shock applied in each period starts at Day 1 and ends at Day 10. More specifically, the accumulated generalized impulse response is the accumulated sum of the impulse responses up to 10 days. Fig.4 translates graphically the accumulated generalized impulse response functions in Table 5 for 10 days. There are four panels in Table 5. The first column of each panel shows the number of days ahead for which impulse responses are added up. Columns 2 to 5 show the responses of each volatility index (VIX, VXN, VDAX and VSTOXX) to a unit step shock in the volatility indexes for 10 days ahead.

The accumulated impulse responses of VIX to a one standard deviation shock in the

innovations of VIX, VXN, VDAX and VSTOXX are provided in Panel A of Table 5. A unit shock is applied to VIX, VXN, VDAX, and VSTOXX, and the corresponding impulse responses of VIX on Day 1 are traced. A unit shock in each volatility index VIX, VXN, VDAX and VSTOXX induces an increase in VIX equal to 1.223285 units, 0.935473 units, 0.660797 units and 0.620181 units, respectively. Similarly, the same unit shock is applied to each volatility index and accumulated responses of VIX on Day 2 are captured. As can be seen, the effects are positive. However, slight decreases from their previous levels are noticed on Day 1, and a similar decrease is observed on Day 3 to the same unit shock. On Day 4, the accumulated impulse responses of VIX to the same unit shock in each volatility index induce an increase in VIX. From Day 4 onward up to Day 10, the effect of the same shock each time gradually dies out and thus induces no more change in the VIX index.

The accumulated impulse responses of VXN to a one standard deviation shock in the innovations of VIX, VXN, VDAX and VSTOXX are presented in Panel B of Table 5. A unit shock is applied to VIX, VXN, VDAX, and VSTOXX, and the impulse responses of VXN on Day 1 are traced. For example, a unit shock in each volatility index VIX, VXN, VDAX and VSTOXX induces an increase in VXN equal to 1.016447 units, 1.329171 units, 0.617268 units and 0.578178 units, respectively. A similar unit shock is applied to each volatility index and accumulated responses of VXN on Day 2 are captured. Surprisingly, VIX, VDAX and VSTOXX induce accumulated increases in the VXN index; increases of 1.036213, 0.685166 and 0.655583 units being observed, respectively. However, the affect on VXN to the same unit shock in volatility indexes induces a slight decrease from previous levels on Day 3, while on Day 4 the effect increases to a similar shock. This effect is almost negligible after Day 4 up to Day 10, although there is noticeable persistence.

Table 5

Accumulated generalized impulse responses to one standard deviation shock in each volatility index.

<i>Panel A: Accumulated Response of ΔVIX.</i>					<i>Panel B: Accumulated response of ΔVXN.</i>				
Days	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$	Days	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$
1	1.223285 (0.01888)	0.935473 (0.02350)	0.660797 (0.02470)	0.620181 (0.02547)	1	1.016447 (0.02586)	1.329171 (0.02101)	0.617268 (0.02870)	0.578178 (0.02767)
2	1.104341 (0.03342)	0.825438 (0.03592)	0.642341 (0.03564)	0.616088 (0.03613)	2	1.036213 (0.04036)	1.307864 (0.03772)	0.685166 (0.04368)	0.655583 (0.04182)
3	1.022642 (0.03880)	0.754765 (0.04249)	0.547642 (0.04017)	0.538585 (0.04084)	3	0.990511 (0.04707)	1.224273 (0.04661)	0.648563 (0.04951)	0.633493 (0.04911)
4	1.035240 (0.03468)	0.768145 (0.03934)	0.558477 (0.03691)	0.538734 (0.03732)	4	0.996540 (0.04500)	1.232274 (0.04506)	0.639237 (0.04691)	0.617312 (0.04639)
5	1.031482 (0.03299)	0.765117 (0.03740)	0.564597 (0.03575)	0.544800 (0.03609)	5	0.991283 (0.04326)	1.230699 (0.04296)	0.639768 (0.04556)	0.617337 (0.04492)
8	1.034025 (0.03305)	0.767289 (0.03761)	0.563485 (0.03575)	0.544937 (0.03614)	8	0.991584 (0.04299)	1.231206 (0.04288)	0.640413 (0.04547)	0.618872 (0.04482)
10	1.034141 (0.03302)	0.767513 (0.03760)	0.563327 (0.03571)	0.544905 (0.03610)	10	0.991369 (0.04295)	1.231131 (0.04287)	0.640148 (0.04545)	0.618737 (0.04480)
<i>Panel C: Accumulated response of $\Delta VDAX$.</i>					<i>Panel D: Accumulated response of $\Delta VSTOXX$.</i>				
Days	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$	Days	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$
1	0.729594 (0.02721)	0.627240 (0.02806)	1.350645 (0.02123)	1.217041 (0.02311)	1	0.741438 (0.03043)	0.636157 (0.02963)	1.317797 (0.02496)	1.462461 (0.02282)
2	1.008327 (0.04072)	0.797596 (0.04272)	1.313461 (0.03741)	1.227502 (0.03824)	2	1.099532 (0.04405)	0.882780 (0.04625)	1.305453 (0.04313)	1.388541 (0.04218)
3	1.035611 (0.04871)	0.802762 (0.05274)	1.253509 (0.04693)	1.181922 (0.04818)	3	1.104198 (0.05290)	0.834173 (0.05671)	1.259573 (0.05175)	1.345970 (0.05167)
4	0.948651 (0.04681)	0.737734 (0.05193)	1.233968 (0.04436)	1.151739 (0.04545)	4	0.995558 (0.05041)	0.763096 (0.05538)	1.219730 (0.04895)	1.322224 (0.04853)
5	0.947298 (0.04457)	0.737515 (0.04938)	1.231428 (0.04238)	1.151135 (0.04349)	5	1.008533 (0.04775)	0.778895 (0.05250)	1.219202 (0.04699)	1.315021 (0.04660)
8	0.957586 (0.04399)	0.746294 (0.04892)	1.237031 (0.04196)	1.156736 (0.04306)	8	1.016445 (0.04724)	0.782988 (0.05208)	1.227248 (0.04646)	1.322970 (0.04603)
10	0.957744 (0.04391)	0.746536 (0.04887)	1.237102 (0.04194)	1.156721 (0.04302)	10	1.016005 (0.04718)	0.782458 (0.05205)	1.227569 (0.04642)	1.323179 (0.04598)

Generalized impulses, standard errors(Monte Carlo).

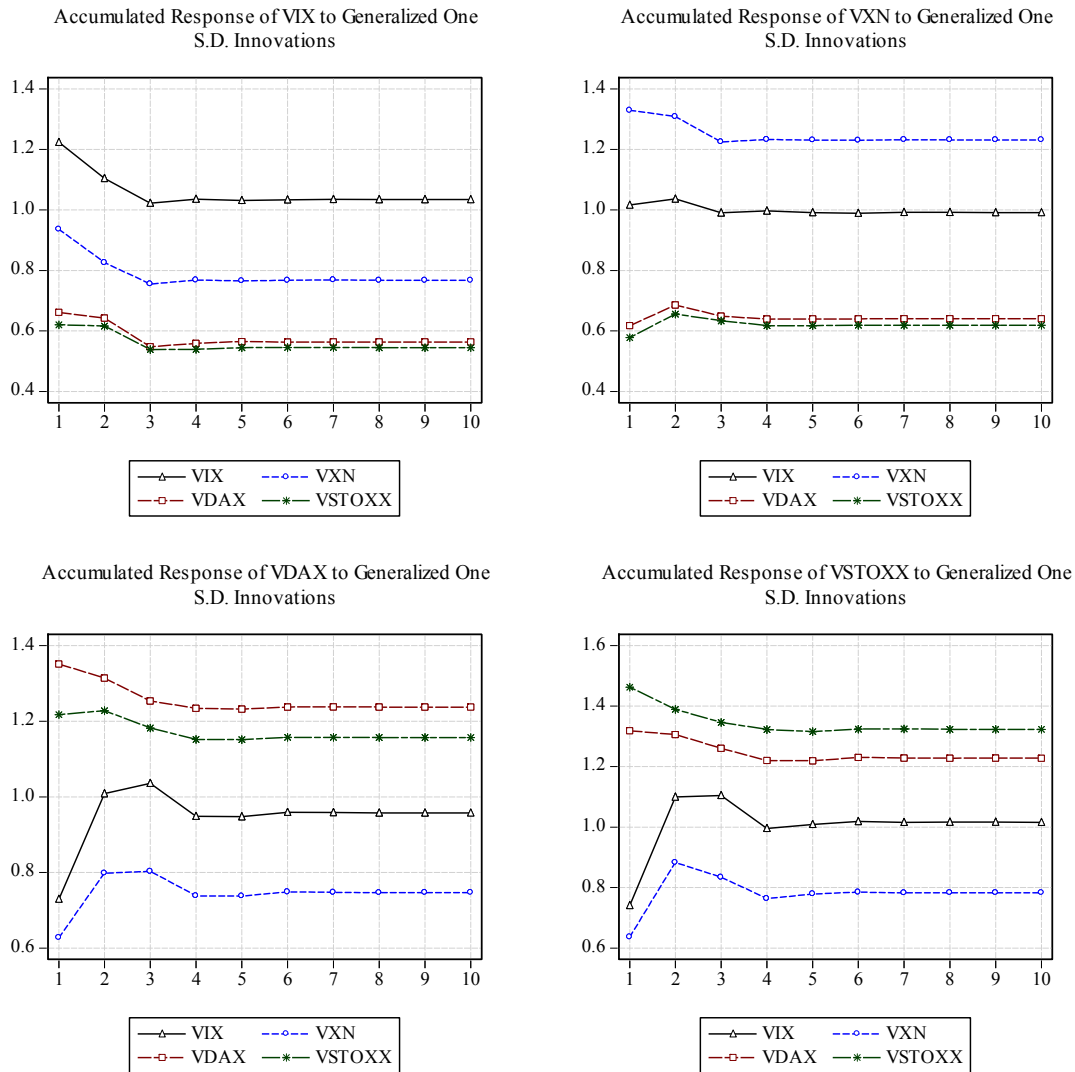


Fig. 4. Accumulated impulse response functions of VIX, VXN, VDAX and VSTOXX.

The accumulated impulse responses of the VDAX volatility index to a one standard deviation shock in the VIX, VXN, VDAX and VSTOXX volatility indexes are presented in Panel C of Table 5. As can be seen on Day 1 impulse responses of VDAX to a unit shock in each VIX, VXN, VDAX and VSTOXX likewise induce an increase in VDAX, increases from the previous day being 0.729594 units, 0.627240 units, 1.350645 units and 1.217041 units, respectively. When a similar unit shock is applied to all four volatility indexes, on Day 2,

considerable increases over the previous period are observed in the VDAX against VIX and VXN, of 1.008327 units and 0.797596 units, respectively. While, VSTOXX marginally increases VDAX, on Day 3, VIX and VXN continue to affect VDAX and increase it further from the previous levels, although the effect of the other two indexes die out. Then on the Day 4, the effect decreases and on Days 5 and 6, VDAX slightly increases to a shock in both VIX and VXN. Finally, the accumulated impulse responses of VSTOXX are reported in Panel D of Table 5. As can be seen, the impulse responses for 10 days exhibit almost similar patterns to that of the VDAX impulse responses, which are further graphically confirmed in Figure 4.

We conclude from the results presented in Table 5 that all volatility indexes respond strongly to a shock in the VIX index, which is contemporaneous (Day 1) and which continues for about 4 days in the US volatility indexes (VIX and VXN), with the effects dying out afterwards. Second, the responses from the European volatility indexes—VDAX and VSTOXX—are affected by the shocks in the US volatility indexes—VIX and VXN—contemporaneously and up to 6 days. Finally, the response from the US volatility indexes are found to the shocks in the European volatility indexes contemporaneously up to Day 2, the effect decreasing considerably up to Day 10. Hence, the results in Table 5 reconcile to our previous results of correlation analysis and Granger causality tests.

5.2.2. Variance Decompositions

The impulses can be utilized in the derivation of the forecast error variance decompositions. As impulse response functions capture the effects of a unit shock to one endogenous variable onto the other variables in the dynamic VAR system, variance decomposition separates the variances in an endogenous variable into the component shocks to the dynamic VAR. Thus, variance decomposition analysis is important in providing information about the relative

importance of each random innovation in affecting the variables in the dynamic VAR system, and more specifically, to determine how important innovations of all endogenous variables in the system are in order to forecast error variances of each variable for the specified number of steps ahead (see, e.g., Koop et al., 1996; Pesaran and Shin, 1998; Mills, 1999).

Mathematically, variance decomposition is defined as the proportion of the n-step ahead forecast error variance of variable i that is accounted for by innovations in variable z_j in the dynamic VAR system. Representing the forecast error variance decompositions by $\sigma_{ij}(n)$, for n steps, i.e., $n=0, 1, 2, 3, \dots$,

$$\sigma_{ij}(n) = \frac{\sum_{l=0}^n (e_i' \Psi_l P e_j)^2}{\sum_{l=0}^n (e_i' \Psi_l \Sigma \Psi_l' e_i)}, \quad i, j = 1, \dots, m. \quad (9)$$

Table 6 provides the variance decompositions for VIX, VXN, VDAX and VSTOXX, which are translated into graphical form in Fig.5. There are four panels in Table 6; the first column of each panel shows the number of days ahead error variances; and the second column reports standard errors, i.e., the forecast error of the variable at a particular forecast horizon. This forecast error is the variation in the current and future expected values of the innovations to each variable in the dynamic VAR system. The remaining four columns in each panel of Table 6 provide the percentage forecast error variances due to specific innovations, each row adding up to 100 percent.

Table 6Variance decomposition of ΔVIX , ΔVXN , $\Delta VDAX$, and $\Delta VSTOXX$.

<i>Panel A: Variance decomposition of ΔVIX.</i>						<i>Panel B: Variance decomposition of ΔVXN.</i>					
Days	Standard Errors.	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$	Days	Standard Errors.	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$
1	1.223285	100.0000	0.000000	0.000000	0.000000	1	1.329171	58.48003	41.51997	0.000000	0.000000
2	1.231291	99.63699	0.057816	0.217906	0.087292	2	1.333129	58.15535	41.45363	0.305416	0.085612
3	1.235645	99.37323	0.068002	0.445518	0.113254	3	1.336345	57.99272	41.57343	0.306730	0.127126
4	1.235938	99.33643	0.070182	0.446493	0.146897	4	1.336590	57.97346	41.55971	0.320099	0.146733
5	1.235984	99.33005	0.070180	0.452684	0.147086	5	1.336611	57.97319	41.55922	0.320840	0.146756
8	1.236002	99.32763	0.070234	0.453438	0.148700	8	1.336621	57.97282	41.55868	0.320878	0.147626
10	1.236002	99.32761	0.070236	0.453442	0.148715	10	1.336621	57.97281	41.55867	0.320879	0.147634
<i>Panel C: Variance decomposition of $\Delta VDAX$.</i>						<i>Panel D: Variance decomposition of $\Delta VSTOXX$.</i>					
Days	Standard Errors.	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$	Days	Standard Errors.	ΔVIX	ΔVXN	$\Delta VDAX$	$\Delta VSTOXX$
1	1.350645	29.17970	0.634103	70.18620	0.000000	1	1.462461	25.70287	0.538667	55.01195	18.74651
2	1.400187	31.11412	0.815031	67.72677	0.344080	2	1.534841	28.77918	0.564808	52.42384	18.23217
3	1.403441	31.00779	0.841391	67.79588	0.354944	3	1.537804	28.66931	0.839876	52.32806	18.16276
4	1.406715	31.24575	0.837742	67.53412	0.382388	4	1.542257	29.00020	0.849618	52.04437	18.10581
5	1.406724	31.24547	0.837813	67.53357	0.383144	5	1.542458	28.99972	0.852893	52.03488	18.11251
8	1.406780	31.25037	0.838094	67.52829	0.383247	8	1.542524	29.00215	0.853356	52.03229	18.11220
10	1.406780	31.25037	0.838095	67.52828	0.383248	10	1.542525	29.00214	0.853360	52.03228	18.11222

Standard errors: Monte Carlo method.

The results of the variance decomposition for VIX are reported in Panel A of Table 6. VIX explains all of its own 1-day ahead forecast error variance and 99.327% of its 10-day ahead forecast error variance. However, none of the other volatility indexes are able to explain even 1% of the forecast error variance in VIX. The results of the variance decomposition for VXN are presented in Panel B of Table 6. As can be seen, VXN explains 41.519% of its own 1-day ahead forecast error variance and almost similar values for 10-day ahead forecast error variance, likewise, 1-day and 10-day ahead horizons, VIX explains 58.480% and 57.972% of the forecast error variance in VXN, respectively. However, the other two volatility indexes are unable to explain even 1% of the forecast error variance in VXN. The results of the variance decomposition for the VDAX volatility index are displayed in Panel C of Table 6. As can be seen, VDAX explains 70.186% of its own 1-day ahead forecast error variance and 67.528% of its 10-day ahead forecast error variance, while VIX explains 27.179% for 1-day ahead, and 31.250% for 10-day ahead forecast error variance in the VDAX index. None of the other volatility indexes significantly explain the forecast error variance in VDAX. Finally, the result of the variance decomposition for the VSTOXX volatility index is shown in Panel D of Table 6. As can be seen, VSTOXX explains 18.746% of its own 1-day ahead forecast error variance and analogous values for 10-day ahead forecast error variance. Similarly, at the 1-day ahead time horizon, VIX and VDAX explain 25.702% and 55.011%, respectively, of the forecast error variance in VSTOXX. For the 10-day ahead time horizon, VIX and VDAX explain 29.002% and 52.032%, respectively, of the forecast error variance in VSTOXX.

The main conclusion drawn from Table 6 is that the VIX volatility index can explain on average 58%, 31% and 29% of the forecast error variances of VXN, VDAX and VSTOXX, respectively. On the other hand, VDAX is the second most influential volatility index, particularly in the European context which can explain on average 52% of the forecast error

variance of VSTOXX. The results of Table 6 reconcile to the results from impulse response functions.

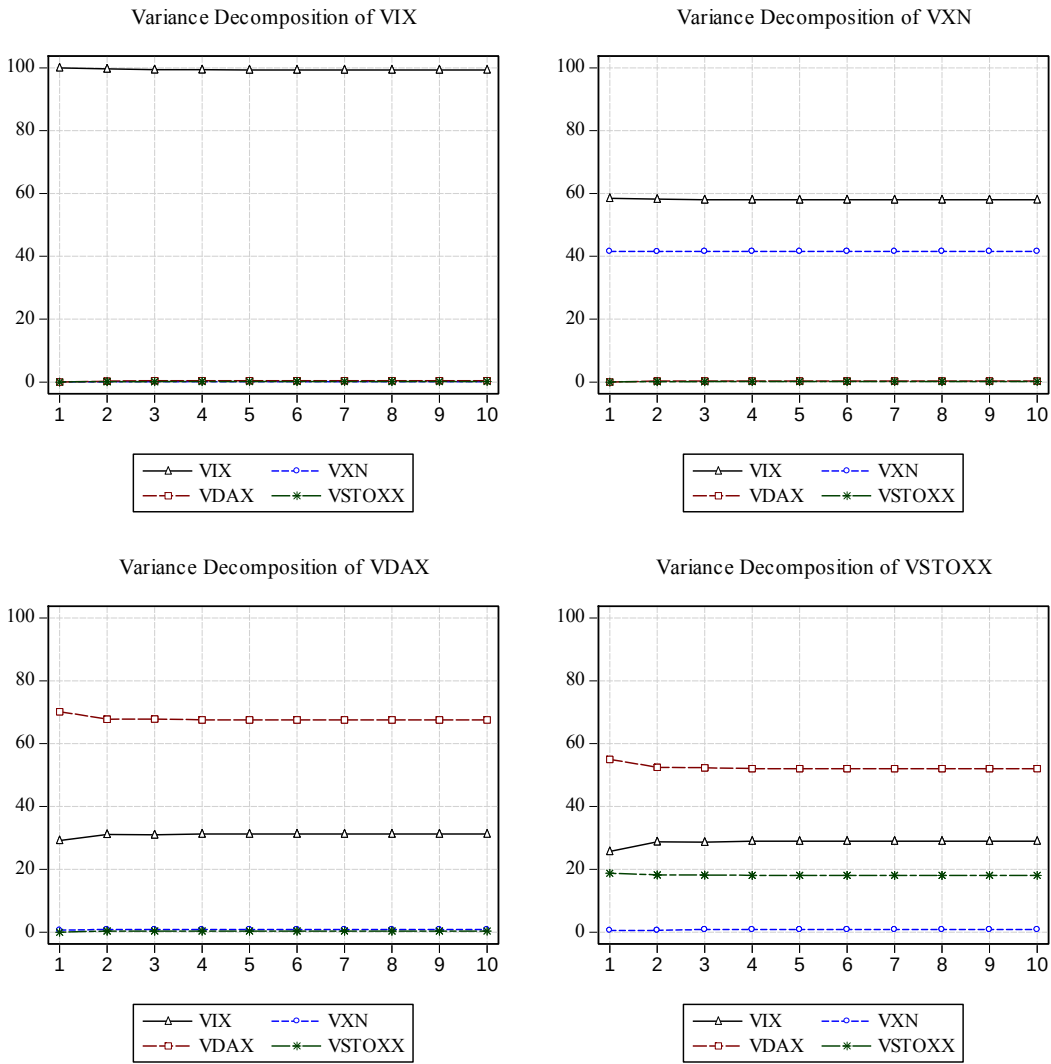


Fig. 5. Variance decomposition of changes in VIX, VXN, VDAX and VSTOXX.

6. Conclusion

We investigated the asymmetric return-volatility phenomenon in the newly adapted robust volatility indexes, i.e., VIX, VXN, VDAX, and VSTOXX. Bollen and Whaley (2004) documented that the net buying pressure for stock index put options from institutional investors' seeking to hedge their portfolios induces increases in implied volatilities. Likewise, new implied volatility indexes incorporate both OTM put and call options, are thus highly informed and robust measures. Accordingly, they should present more pronounced asymmetric return-volatility relationships in comparison to their old counterparts. Further, we examined the dynamic implied volatility transmissions across volatility indexes using techniques such as Granger causality, generalized impulse response function and variance decomposition.

There is noticeable evidence that implied volatility indexes VIX, VXN, VDAX, and VSTOXX from February 2001 through June 2008 respond in a pronounced asymmetric fashion to the negative and positive returns of their corresponding stock indexes. VIX presents the highest asymmetry, followed by VSTOXX, VDAX and VXN volatility indexes, respectively. These volatility indexes sharply rise in market declines (fear) and fall in market rallies (exuberance). On the other hand, there are significant spillover effects across volatility indexes, bi-directional causality running between the volatility indexes. A shock to the VIX index increases considerably the levels of the other volatility indexes—VXN, VDAX and VSTOXX—in a contemporaneous manner and the affect of the shock persists on average for 4 to 6 days. While in the European context, VDAX is the dominant source of information. Therefore, a shock to VDAX increases the level of VSTOXX. Conversely, a shock to VDAX to some extent affects the levels of US volatility indexes VIX and VXN as well. Moreover, in variance decompositions, the VIX volatility index can explain on average 58%, 31% and 29% of the forecast error variances for VXN, VDAX and VSTOXX, respectively, while

VDAX can explain 52% of the forecast error variance for VSTOXX.

Our results have a number of implications: First, as we found that newly adapted volatility indexes are highly negatively correlated with their corresponding stock indexes. The new volatility indexes are important instruments for hedging stock portfolios. Derivatives exchanges provide liquid markets for futures and options underlying these volatility indexes. Therefore, the position in futures or options on a volatility index can more accurately hedge the stock portfolio position without considering complicated stock index option trading strategies. Second, when the stock index drops, then the volatility index sharply rises. Therefore, new volatility indexes are useful not only for assessing potential risks, but also for speculative transactions by risk-seeking investors. Third, since the new volatility indexes are based on the robust MF concept and provide better tradability, it is easier for derivatives issuers to engineer structured products on the volatility indexes. Fourth, trading strategies in range could generate profits, for instance, a volatility long position in decreasing volatility markets while a volatility short position in increasing volatility markets. Fifth, we found high correlation between the implied volatility indexes, indicating that investors' expectations about the future realized volatilities is robustly related across the Atlantic. Therefore, portfolio managers and option traders could incorporate correlated expectations using the volatility indexes' implied correlations into their models. Finally, implied correlations from these volatility indexes can be incorporated into value-at-risk estimations.

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